

Tarea: Construcción del Dataset Integrado (Demografía, Salud y Economía)

Objetivo de la Tarea:

Completar la construcción del dataset agregando las variables `percentage_health` y `gdp` al dataset demográfico creado en clase.

La salida final debe contener **únicamente las 8 columnas requeridas**:

1. `entity`
2. `year`
3. `population`
4. `births`
5. `deaths`
6. `child_deaths_under_5`
7. `percentage_health`
8. `gdp`

```
In [ ]: import pandas as pd
import numpy as np

# Configuración de headers para Our World in Data
headers = {'User-Agent': 'Our World In Data data fetch/1.0'}
```

PARTE 1. Dataset Base Creado en Clase (Demografía UN WPP)

Cargamos y estructuramos las 6 variables base:

- `entity` y `year` (País y Año)
- `population` (Población total estimada)
- `births` (Nacimientos anuales)
- `deaths` (Defunciones anuales)
- `child_deaths_under_5` (Muertes de niños menores a 5 años)

```
In [ ]: # 1. Nacimientos y Defunciones
url_bd = "https://ourworldindata.org/grapher/births-and-deaths-projected-to-2100.cs"
df_bd = pd.read_csv(url_bd, storage_options=headers)
df_bd = df_bd.rename(columns={
    'births_sex_all_age_all_variant_estimates': 'births',
    'deaths_sex_all_age_all_variant_estimates': 'deaths'
})[['entity', 'code', 'year', 'births', 'deaths']]
```

```
# 2. Población
url_pop = "https://ourworldindata.org/grapher/population-with-un-projections.csv?v=
df_pop = pd.read_csv(url_pop, storage_options=headers)
df_pop = df_pop.rename(columns={
    'population__sex_all__age_all__variant_estimates': 'population'
})[['entity', 'code', 'year', 'population']]

# 3. Muertes en menores de 5 años (child_deaths_under_5)
url_cd = "https://ourworldindata.org/grapher/number-of-child-deaths-unwpp.csv?v=1&c
df_cd = pd.read_csv(url_cd, storage_options=headers)
df_cd = df_cd.rename(columns={
    'deaths__sex_all__age_0_4__variant_estimates': 'child_deaths_under_5'
})[['entity', 'code', 'year', 'child_deaths_under_5']]

# Unir el dataset base de la clase
df_clase = df_pop.merge(df_bd, on=['entity', 'code', 'year'], how='inner')
df_clase = df_clase.merge(df_cd, on=['entity', 'code', 'year'], how='inner')

print("Dimensiones del dataset de clase:", df_clase.shape)
df_clase.head(3)
```

Dimensiones del dataset de clase: (18942, 7)

```
Out[ ]:
```

	entity	code	year	population	births	deaths	child_deaths_under_5
0	Afghanistan	AFG	1950	7776180.0	383985.0	290972.0	160791
1	Afghanistan	AFG	1951	7879343.0	391002.0	288752.0	158566
2	Afghanistan	AFG	1952	7987784.0	397663.0	288059.0	157832

PARTE 2. Extracción de los Nuevos Indicadores Solicitados

2.1 Indicador: Total healthcare spending as a share of GDP (2000 to 2023)

Extraemos la variable `percentage_health`.

```
In [ ]: # Carga de Total healthcare spending as a share of GDP
url_health = "https://ourworldindata.org/grapher/total-healthcare-expenditure-gdp.c
df_health = pd.read_csv(url_health, storage_options=headers)

# Renombrar la columna a 'percentage_health' como lo pide la tarea
df_health = df_health.rename(columns={
    'current_health_expenditure__che__as_percentage_of_gross_domestic_product__gdp__
'})

print("df_health shape:", df_health.shape)
df_health[['entity', 'year', 'percentage_health']].head(3)
```

df_health shape: (4756, 4)

```
Out[ ]:
```

	entity	year	percentage_health
0	Afghanistan	2002	9.443391
1	Afghanistan	2003	8.941258
2	Afghanistan	2004	9.808474

2.2 Indicador: GDP per capita

Extraemos la variable `gdp`.

```
In [ ]: # Carga de GDP per capita
url_gdp = "https://ourworldindata.org/grapher/gdp-per-capita-worldbank.csv?v=1&csvT
df_gdp = pd.read_csv(url_gdp, storage_options=headers)

# Renombrar la columna a 'gdp'
df_gdp = df_gdp.rename(columns={
    'ny_gdp_pcap_pp_kd': 'gdp'
})

print("df_gdp shape:", df_gdp.shape)
df_gdp[['entity', 'year', 'gdp']].head(3)
```

df_gdp shape: (7445, 5)

```
Out[ ]:
```

	entity	year	gdp
0	Afghanistan	2000	1617.8264
1	Afghanistan	2001	1454.1108
2	Afghanistan	2002	1774.3087

PARTE 3. Construcción del Dataset Final con las 8 Columnas

Unimos las variables `percentage_health` y `gdp` al dataset creado en clase mediante `pd.merge` por `entity` y `year`.

```
In [ ]: # Merge con df_health y df_gdp
df_completo = df_clase.merge(
    df_health[['entity', 'year', 'percentage_health']],
    on=['entity', 'year'],
    how='inner'
).merge(
    df_gdp[['entity', 'year', 'gdp']],
    on=['entity', 'year'],
    how='inner'
)

# Definir la lista exacta de las 8 columnas requeridas por el profesor
```

```
columnas_salida = [  
    'entity',  
    'year',  
    'population',  
    'births',  
    'deaths',  
    'child_deaths_under_5',  
    'percentage_health',  
    'gdp'  
]  
  
# Filtrar y ordenar las columnas requeridas  
df_final = df_completo[columnas_salida].copy()  
  
# Ordenar por país y año  
df_final = df_final.sort_values(by=['entity', 'year']).reset_index(drop=True)  
  
print("¡Dataset completado con éxito!")  
print(f"Dimensiones finales: {df_final.shape[0]} filas y {df_final.shape[1]} columnas")  
print("\nLista de columnas:")  
for i, col in enumerate(df_final.columns, 1):  
    print(f"{i}. {col}")
```

¡Dataset completado con éxito!
Dimensiones finales: 4428 filas y 8 columnas

Lista de columnas:

1. entity
2. year
3. population
4. births
5. deaths
6. child_deaths_under_5
7. percentage_health
8. gdp

PARTE 4. Inspección del Dataset Final

```
In [ ]: # Primeras filas del dataset final  
df_final.head(10)
```

Out []:

	entity	year	population	births	deaths	child_deaths_under_5	percentage_he
0	Afghanistan	2002	21378123.0	1015202.0	239161.0	123215	9.443
1	Afghanistan	2003	22733053.0	1069727.0	243573.0	123894	8.941
2	Afghanistan	2004	23560656.0	1095372.0	244435.0	122315	9.808
3	Afghanistan	2005	24404575.0	1097038.0	241932.0	118420	9.948
4	Afghanistan	2006	25424100.0	1120880.0	245586.0	116622	10.622
5	Afghanistan	2007	25909852.0	1135656.0	245809.0	112800	9.904
6	Afghanistan	2008	26482631.0	1126525.0	235284.0	106511	10.256
7	Afghanistan	2009	27466102.0	1158017.0	237450.0	106480	9.818
8	Afghanistan	2010	28284088.0	1181113.0	237640.0	104547	8.569
9	Afghanistan	2011	29347708.0	1202456.0	235724.0	101679	8.561

In []:

```
# Últimas filas del dataset final
df_final.tail(10)
```

Out []:

	entity	year	population	births	deaths	child_deaths_under_5	percentage_h
4418	Zimbabwe	2014	14207367.0	503038.0	136688.0	32501	8.13
4419	Zimbabwe	2015	14399008.0	488386.0	132188.0	30095	7.45
4420	Zimbabwe	2016	14600297.0	477484.0	128714.0	28013	7.25
4421	Zimbabwe	2017	14812484.0	471228.0	127191.0	26491	6.17
4422	Zimbabwe	2018	15034457.0	470991.0	125092.0	25211	4.67
4423	Zimbabwe	2019	15271377.0	475267.0	126468.0	24647	3.23
4424	Zimbabwe	2020	15526887.0	481152.0	126365.0	23828	2.95
4425	Zimbabwe	2021	15797220.0	488642.0	138738.0	23391	2.67
4426	Zimbabwe	2022	16069061.0	496240.0	124995.0	23249	3.39
4427	Zimbabwe	2023	16340829.0	496917.0	124412.0	22505	2.92

In []:

```
# Tipos de datos e información general
df_final.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4428 entries, 0 to 4427
Data columns (total 8 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   entity                4428 non-null   str
 1   year                  4428 non-null   int64
 2   population            4428 non-null   float64
 3   births                4428 non-null   float64
 4   deaths                4428 non-null   float64
 5   child_deaths_under_5  4428 non-null   int64
 6   percentage_health     4428 non-null   float64
 7   gdp                   4428 non-null   float64
dtypes: float64(5), int64(2), str(1)
memory usage: 276.9 KB
```

Ejemplo: Consulta para México (2000 a 2023)

```
In [ ]: # Visualización de los datos correspondientes a México
df_mexico = df_final[df_final['entity'] == 'Mexico']
df_mexico
```

Out[]:

	entity	year	population	births	deaths	child_deaths_under_5	percentage_h
2545	Mexico	2000	98625553.0	2385977.0	540567.0	66928	4.24
2546	Mexico	2001	100099101.0	2380261.0	543803.0	64012	4.58
2547	Mexico	2002	101548627.0	2374140.0	553391.0	61210	4.83
2548	Mexico	2003	102978517.0	2351177.0	563140.0	58280	5.53
2549	Mexico	2004	104394131.0	2335226.0	575925.0	55625	5.68
2550	Mexico	2005	105811504.0	2318498.0	585859.0	53204	5.58
2551	Mexico	2006	107253666.0	2303997.0	597892.0	50857	5.40
2552	Mexico	2007	108774356.0	2296268.0	611318.0	48847	5.50
2553	Mexico	2008	110374286.0	2297950.0	627177.0	47137	5.44
2554	Mexico	2009	111999720.0	2303532.0	642292.0	45610	5.84
2555	Mexico	2010	113623898.0	2310124.0	651368.0	44196	5.48
2556	Mexico	2011	115243503.0	2315793.0	656666.0	42843	5.30
2557	Mexico	2012	116818208.0	2319451.0	662847.0	41517	5.42
2558	Mexico	2013	118343572.0	2314458.0	671786.0	40118	5.57
2559	Mexico	2014	119784265.0	2270937.0	683362.0	38277	5.35
2560	Mexico	2015	121072310.0	2206765.0	698389.0	36172	5.52
2561	Mexico	2016	122251344.0	2162620.0	717395.0	34306	5.38
2562	Mexico	2017	123400052.0	2124252.0	740091.0	32584	5.31
2563	Mexico	2018	124573713.0	2122316.0	756310.0	31321	5.23
2564	Mexico	2019	125762978.0	2122565.0	765037.0	30194	5.30
2565	Mexico	2020	126799053.0	2100438.0	1068730.0	29075	6.05
2566	Mexico	2021	127648151.0	2087487.0	1150760.0	27958	5.87
2567	Mexico	2022	128613116.0	2063075.0	838627.0	28309	5.70
2568	Mexico	2023	129739761.0	2037689.0	799360.0	25454	5.50

PARTE 5. Exportar el Dataset Resultante

Guardamos el dataset final como archivo CSV para entrega o uso posterior.

```
In [ ]: # Guardar el dataset completado en un archivo CSV
nombre_archivo_csv = "dataset_health_gdp_demografia_final.csv"
```

```
df_final.to_csv(nombre_archivo_csv, index=False)  
print(f"Archivo guardado exitosamente: {nombre_archivo_csv}")
```

Archivo guardado exitosamente: dataset_health_gdp_demografia_final.csv